AI agents turn our fatigue tests into a lean openBIS data structure

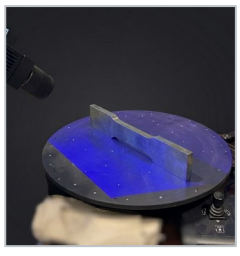
[Authors] · BAM, 7.2 Buildings and Structures

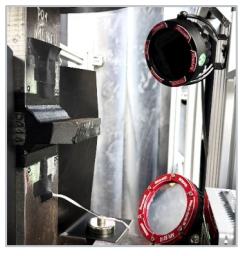
BAM
Bundesanstalt für Materialforschung und -prüfung

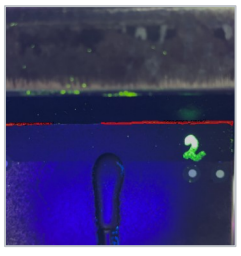
Idea Describe the experiment once, in domain language. AI agents check it against the BAM data standard and the live BAM Data Store, design a flat structure and create only what is really missing. Humans approve every decision.

1 The tests: S-N curves of welded joints

Butt and cruciform joints, randomised load levels – scatter starts before the test, so every specimen must be traceable.


3D scan


test rig


crack monitoring


fracture surface

01–02 Preparation — quality check, grinding
03–06 Characterisation — 3D scan, weld rating, geometry
07–11 Test rig — gauges, cyclic test, evaluation
12 Post-test — fracture surface

Per specimen: 34 data items · 10 instruments · 17 file kinds

2 The agent approach: check first

Workflow 12 steps, domain language
BAM standard bam-masterdata, 141 object types
BAM Data Store what exists? read-only

AI agents + scripts
understand · design · compare – every element gets one verdict

20	2	41	40
REUSE	EXTEND	CREATE	LIVE-ONLY
use as is	add terms	define new	migrate

Humans approve twice (design, gap report). Only missing types are created; new types go to the BAM standard as a pull request.

3 Seven steps: scripts count, agents decide, humans approve

1 Catalog the standard → 2 Inventory the instance → 3 Read the workflow → 4 Design a flat structure → 5 Gap analysis (4 verdicts) → 6 Python classes, import-tested → 7 Apply to openBIS, dry run first

■ AI agent ■ deterministic script

✔ human approval (after 4) ✔ human approval (after 5)

4 Result: one specimen, three steps, one level

Specimen "Next step" on top · sections 01 · 02 · 04 · 05 · 06

- **SCAN** steps 03–04
- **TEST** steps 07–11
- **FRACTURE** step 12

Instruments & consumables: created once in the inventory, linked as parents

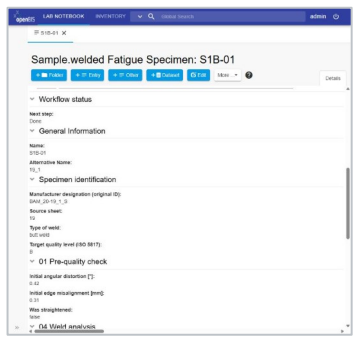
2	6+2	0	1
new object types	vocabularies new + extended	new dataset types	link depth

5 Connected across departments

FB 9.3 welding + process docs → **Specimen** one object, one place → **FB 7.2** 12-step test workflow

Welding and test data meet at the same specimen.

Instruments and datasets are linked once – and found from both sides.



openBIS: specimen with "Next step"

Take-home

Check, don't just generate deterministic catalog, real import test, dry run

Humans decide design & gap report approved; domain calls stay with experts

Reusable pattern any department: describe → agents check → approve

Sicherheit in Technik und Chemie